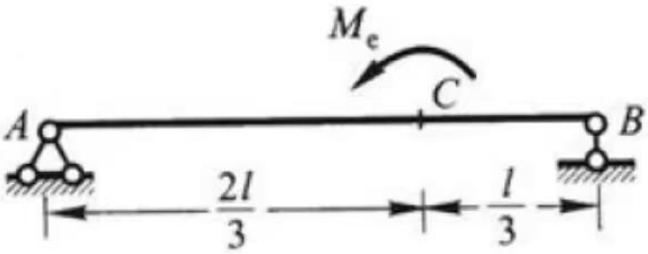
材料为线弹性，弯曲刚度为EI的各梁及其承载情况分别如图所示，不计剪力的影响，试用单位力法求各梁截面A的挠度和转角，以及截面C的挠度。



$F_B \cdot l = M_e$ ∴ $F_B = \frac{M_e}{l}$ $F_A = -F_B = -\frac{M_e}{l}$

对 AC $M_{AC}(x) = -\frac{M_e}{l}\cdot x$ 对 CB. $M_{CB}(x) = \frac{M_e}{l}\cdot x$.

$\omega_{AC} = \frac{1}{2}\cdot\frac{2}{3}l\left(-\frac{2}{3}M_e\right) = -\frac{2}{9}M_e l$. $x_{c1} = \frac{2}{3}\cdot\frac{2}{3}l = \frac{4}{9}l$.

$\omega_{CB} = \frac{1}{2}\cdot\frac{1}{3}l\cdot\left(\frac{2}{3}M_e\right) = \frac{1}{18}M_e l$ $x_{c2} = \frac{2}{3}\cdot\frac{1}{3} = \frac{2}{9}l$

$\Delta_A = 0$.

① 在 A 施加单位力矩 $\bar{M} = 1$. ∴ $\bar{M}_{AC}(x_{c1}) = -\frac{5}{9}$ $\bar{M}_{CB}(x_{c2}) = -\frac{2}{9}$

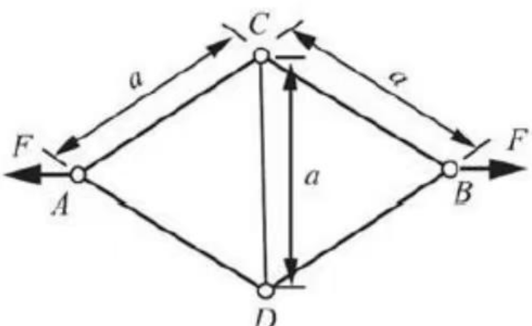
$\theta_A = \frac{1}{EA}\left[\left(-\frac{2}{9}M_e l\right)\times\left(-\frac{5}{9}\right) + \left(\frac{1}{18}M_e l\right)\cdot\left(-\frac{2}{9}\right)\right] = \frac{M_e l}{9EI}$

② 在 C 施加向下单位力 $P = 1$ ∴ $\bar{F}_A = \frac{1}{3}$ $\bar{F}_B = \frac{2}{3}$

$\bar{M}_{AC}(x_{c1}) = \frac{4}{27}l$ $\bar{M}_{BC}(x_{c2}) = \frac{4}{27}l$.

∴ $\Delta_C = \frac{1}{EI}\left[\omega_{AC}\cdot\bar{M}_{AC}(x_{c1}) + \omega_{BC}\cdot\bar{M}_{BC}(x_{c2})\right] = -\frac{2}{81}\frac{M_e l^2}{EI}$.

材料为线弹性，拉压刚度为EA的杆系及其承载情况分别如图所示，试用单位力法求结点A、B间和结点C、D间的相对位移



$+\infty$.

得: $F_{AC}=F_{AD}=F_{DB}=F_{CB}=\frac{\sqrt{3}}{3}F$ $F_{CD}=-\frac{\sqrt{3}}{3}F$.

A.B 施加单位力:

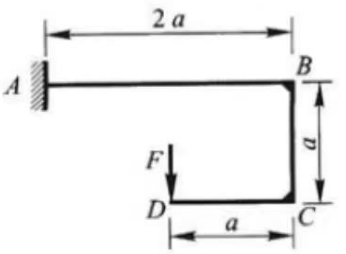
$$\Delta_{AB}=\sum\frac{F_i\bar{F}_i l_i}{E_iA_i}=\frac{\frac{\sqrt{3}}{3}F\cdot\frac{\sqrt{3}}{3}a}{EA}+\frac{(-\frac{\sqrt{3}}{3}F)(-\frac{\sqrt{3}}{3})a}{EA}=\frac{5}{3}\frac{Fa}{EA}$$ 拉伸.

在CD施加单位力.

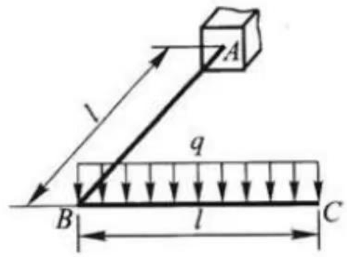
此时 $F_{DB}=F_{AC}=F_{AD}=F_{CB}=0$ $F_{CD}=0$

$\Delta_{CD}=-\frac{\sqrt{3}}{3}F\cdot a/EA=-\frac{\sqrt{3}Fa}{EA}$ 收缩

弯曲刚度为EI，扭转刚度为GIp的圆截面刚架及其承载情况分别如图所示，不计轴力和剪力的影响，试用单位力法求下列指定位移



截面D的线位移和角位移

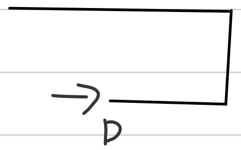


截面C的铅垂位移和角位移。

①

对DC: $M_{DC}(x)=-Fx$ 对CB: $M_{CB}(x)=-Fa$.

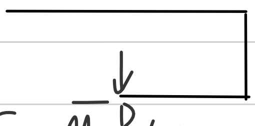
对BA $M_{BA}(x)=Fx-Fa$.



D处施加水平单位力.

$$\begin{cases}\bar{M}_{DC}(x)=0\\ \bar{M}_{CB}(x)=-x\\ \bar{M}_{BA}(x)=-a\end{cases}$$

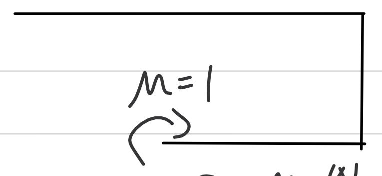
$$\therefore \Delta_{Dx}=\int_0^a\frac{1}{EA}\bar{M}_{DC}(x)M_{DC}(x)dx+\int_0^a\frac{1}{EA}\bar{M}_{CB}(x)M_{CB}(x)dx+\int_0^{2a}\frac{1}{EA}\bar{M}_{BA}(x)M_{BA}(x)dx=\frac{Fa^3}{2EI}$$



在垂直方向施加垂直单位力.

$$\begin{cases}\bar{M}_{DC}(x)=-x\\ \bar{M}_{CB}(x)=-a\\ \bar{M}_{BA}(x)=x-a\end{cases}$$

$$\Delta_{Dy}=\int_0^a\frac{1}{EI}(-x)(-Fx)dx+\int_0^a\frac{1}{EI}(-a)(-Fa)dx+\int_0^{2a}\frac{1}{EI}(x-a)(Fx-Fa)dx=\frac{2Fa^3}{EI}$$



在D施加单位力矩.

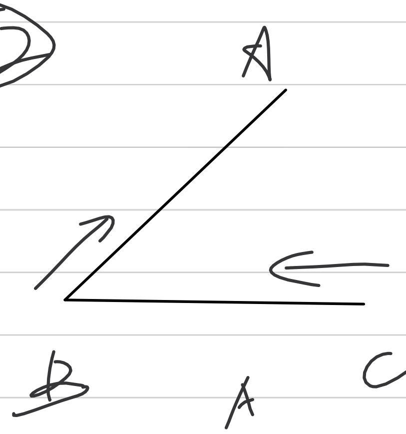
$$\begin{cases} M_{DC}(x)=1 \\ M_{CB}(x)=1 \\ M_{BA}(x)=1 \end{cases}$$

$$\theta_D=\int_0^a \frac{1}{EI}(-Fx)dx+\int_0^a \frac{1}{EI}(-Fa)dx+\int_0^{2a}\frac{1}{EI}(-Fx+Fa)dx$$

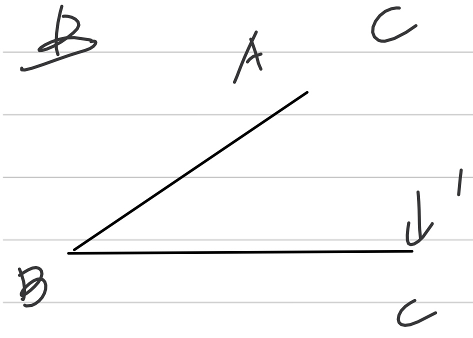
$$=\frac{3Fa^2}{EI}.$$

$$\begin{cases} A_{Dx}=\frac{Fa^3}{2EI} \\ A_{Dy}=\frac{2Fa^3}{EI} \end{cases} \quad \theta_D=\frac{3Fa^2}{EI}$$

②



$$\begin{cases} M_{BC}(x)=\frac{1}{2}qx^2 & T(x)=0 \\ M_{AB}(x)=qlx & T(x)=-\frac{1}{2}ql^2 \end{cases}$$

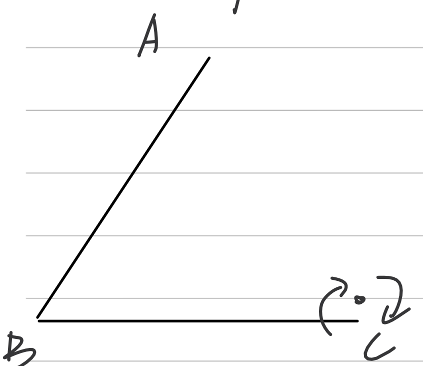


$$\begin{cases} \overline{M}_{BC}(x)=x. & \overline{T}_{BC}(x)=0 \\ \overline{M}_{AB}(x)=x & \overline{T}_{AB}=-l. \end{cases}$$

$$\therefore \Delta_C=\int_0^l \frac{1}{EI}x\frac{1}{2}qx^2dx+\int_0^l\frac{1}{EI}x\,qlx\,dx+\int_0^l\frac{1}{GI_p}(l\cdot\frac{1}{2}ql^2)dx$$

$$=\frac{11}{24}\frac{ql^4}{EI}+\frac{ql^4}{2GI_p}.$$

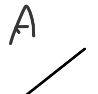
角位移有2个. 一是纸面内的位移. 二是垂直纸面的平面的位移.

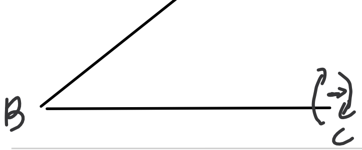


$$\begin{cases} \overline{M}_{BC}(x)=-1 & \overline{T}_{BC}(x)=0 \\ \overline{M}_{AB}(x)=0 & \overline{T}_{AB}(x)=-1 \end{cases}$$

$$\theta_{C1}=\int_0^l\frac{1}{EI}(-\frac{1}{2}qx^2)dx+\int_0^l\frac{1}{GI_p}(-\frac{1}{2}ql^2)dx$$

$$=\frac{ql^3}{6EI}+\frac{ql^3}{2GI_p}$$





$$\begin{cases} \bar{M}_{BC}(x) = 0 & \bar{T}_{BC} = -1 \\ \bar{M}_{AB}(x) = 1 & \bar{T}_{AB} = 0. \end{cases}$$

$$\theta_{c2} = \underbrace{\int_0^l \frac{1}{EI}(-qlx)dx = -\frac{ql^3}{2EI}}$$

综上.

$$\begin{cases} \Delta_c = \frac{11}{24}\frac{ql^4}{EI} + \frac{9}{2}\frac{ql^4}{GI_p} \\ \theta_{c1} = \frac{ql^3}{6EI} + \frac{ql^3}{2GI_p}. \\ \theta_{c2} = \frac{-ql^3}{2EI}. \end{cases}$$